Home-Goods E-commerce: Growth Is Real, but Returns, Channel Quality and Promotions Need Attention

Analytical Report - orders from 1 Jan 2024 to 31 Dec 2025 - prepared 26 October 2026

1. Executive summary

The business sold **$1,881,974 of net product revenue** across 8,456 completed orders in two years, at a **49.2% gross margin** (see caveat on cost data below). Revenue grew **21.2%** in 2025 ($850.8K to $1,031.2K) and was higher than the same month of the prior year in all 12 months. Four findings should shape next quarter's priorities:

1. **Furniture returns are an internal outlier (High priority).** 12.1% of furniture order lines are returned versus 4.6% elsewhere ($p < 0.001$). Furniture is 17.2% of revenue but 33% of all returned value. All 12 furniture products sit at or above 7.8% return rate; only 6 of 78 other products do.
2. **Paid social brings volume, not buyers (High priority, pending spend data).** It supplies 23.1% of registered customers but 12.7% of revenue. Only 44% of its customers ever buy, versus 59%-68% for organic, paid search, email and referral, and the gap holds within each signup year.
3. **2025 growth is acquisition-led (Medium).** Signups rose 81% and active buyers 92%, while revenue per active customer fell 37%; only 50% of 2024 buyers purchased again in 2025.
4. **Promotions dilute margin, and their incremental value is unproven (Medium).** HOLIDAY20 orders earn 37.9% margin versus 50.4% without a code, and the code is used mostly by existing customers.

Context from external sources: US e-commerce grew 5.4% in 2025 [S2], so this business is outgrowing the market, but from a small and quickly expanding base. NRF estimates 19.3% of US online sales are returned [S1]; even furniture at 10.7% (by value) sits below that, so the furniture issue is a *relative* priority, not proof of an absolute returns crisis.

2. Business context and scope

The data describes an online home-goods retailer: 3,600 registered customers across 35 US states, 90 products in 8 categories, 9,207 orders (after removing duplicates) and 15,608 order lines. It supports questions on growth, product mix, returns, customer value, acquisition quality and promotions. It does **not** contain marketing spend, return reasons, cost history, inventory or delivery data, so questions on CAC/ROAS, return causes and supply chain are out of scope (see section 9).

3. Data quality in brief

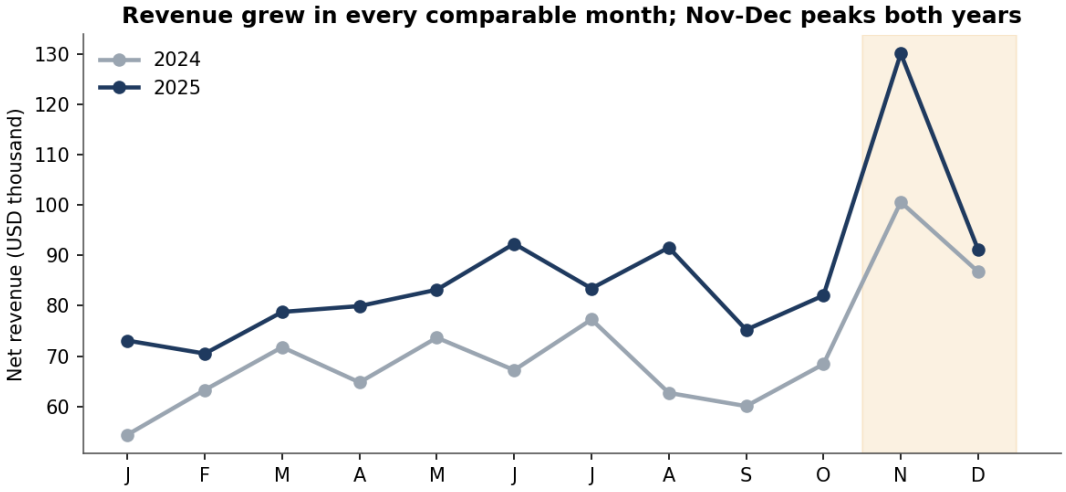
Three issues were treated, one business event was documented, and all 16 post-cleaning validation checks pass: 38 exact duplicate orders removed, 199 missing prices rebuilt from a rule that reproduces 100% of known prices, 24 category labels merged into 8, and a 5% price increase on 2025-01-01 identified as a business event rather than an error. Full detail: `reports/data_quality_report.md`. The SQL layer was run independently on the raw files and reproduces 11 headline metrics to the cent (`python -m src.sql_reconcile`).

4. Methodology and metric definitions

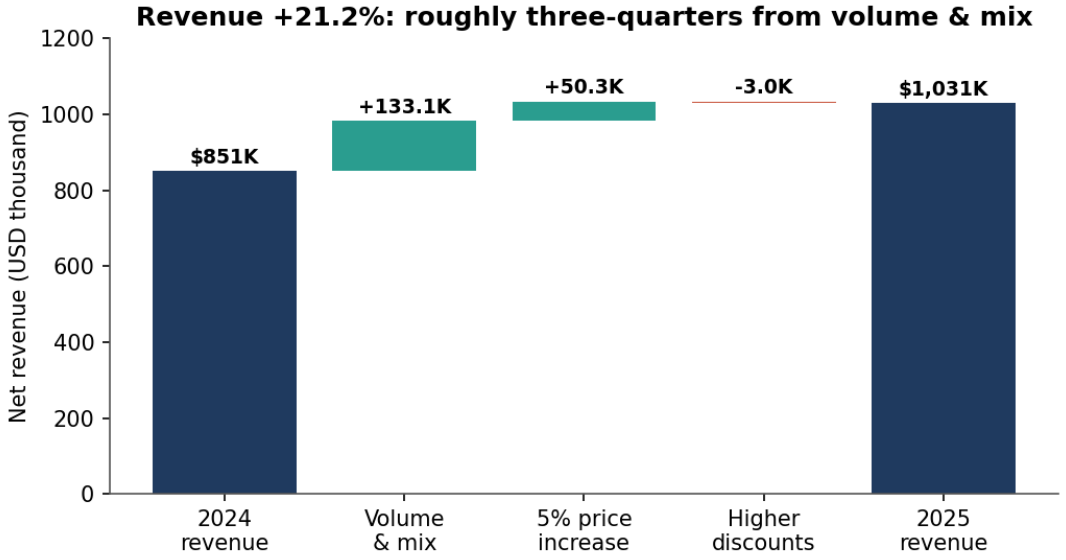
Revenue = net product revenue (quantity x price - discounts) on **completed** orders only. Returned and canceled orders recognise zero revenue; shipping fees ($13,136) are excluded. **Gross profit** = revenue - quantity x unit cost. **Return rate** is measured on order lines in non-canceled orders (returned / (returned + completed)), or by value where stated. Group differences use chi-square tests (alpha = 0.05). Full definitions: `docs/metric_definitions.md`.

5. Key findings

5.1 Revenue grew 21.2%; about three-quarters came from volume and mix, the rest from the 5% price rise



	2024	2025	Change
Net revenue	$850,786	$1,031,188	21.2%
Completed orders	3,867	4,589	18.7%
Units sold	8,215	9,560	16.4%
Average order value	$220.01	$224.71	2.1%
Gross margin	47.8%	50.4%	+2.6 pts

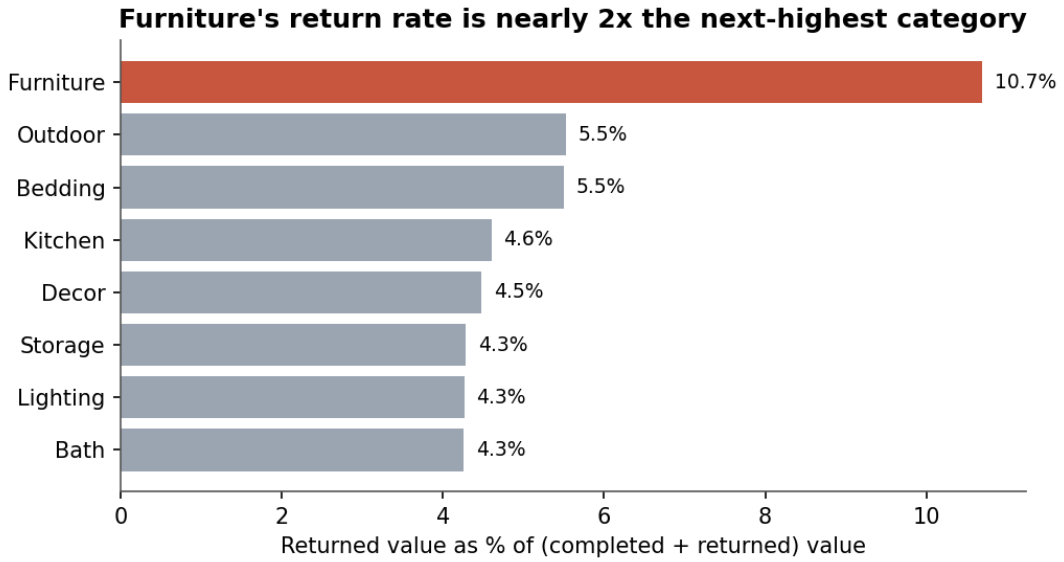


Evidence. Holding 2024 prices constant, 2025 revenue would have been $133.1K higher than 2024 (**74%** of the increase); the price rise added $50.3K (**28%**); higher discount spend took back $3.0K. The three effects sum exactly to the $180.4K difference.

Interpretation. Units grew 16.4% despite the price rise, but the active customer base also nearly doubled (finding 5.5), so no conclusion about price elasticity is possible without a counterfactual.

Caveat on margin. Margin rose 47.8% -> 50.4%, but `unit_cost` is a single static number per product. A 5% price rise on a constant cost produces roughly this change by arithmetic alone; do not read it as operational improvement until cost history is available.

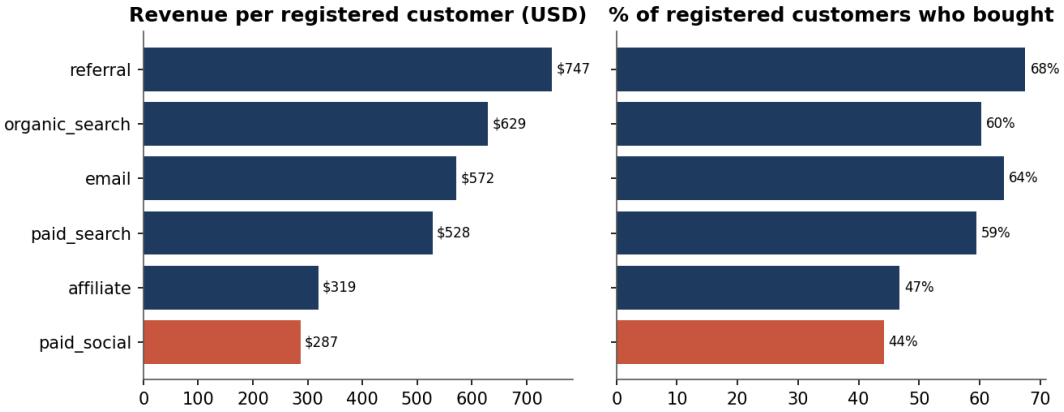
5.2 Furniture returns run at nearly twice the rate of the next-highest category



Insight: Furniture is the largest category ($322.8K, 17.2% of revenue) and has the highest return rate.

- **What the data shows:** 12.1% of furniture lines are returned (95% CI 10.1%-14.1%, n = 1,014) versus 4.6% for all other categories; chi-square p = 1e-25.
- **Evidence:** Consistent in both years (11.6% in 2024, 12.6% in 2025) and after excluding the 81 heavy repeat customers (11.6%). By value, furniture returns total $38.7K: 33% of all returned value. Highest-return products: Stoneware Bar Stool (20.4%), Linen Loveseat (14.6%), Modern Writing Desk (14.3%), Linen TV Stand (13.8%).
- **Why it matters:** *Illustrative scenario (not a forecast):* if furniture returned at the rest-of-catalog rate by value (4.8%), about $21.4K of revenue would have been retained over two years (~1.1% of total revenue; ~$9.9K gross profit at furniture margin). This excludes reverse-shipping, handling and restocking costs, which are not in the data and likely make the true cost larger.
- **Recommended action:** (1) Capture return reason codes for furniture immediately; the data cannot say *why* items come back. (2) Audit the five highest-return products' pages (dimensions, materials, assembly, photos) and carrier damage rates. (3) Pilot fixes on those SKUs first.
- **Priority:** High - strongest statistical evidence in the dataset, concentrated in one category, cheap to diagnose.
- **Measurement:** Furniture line return rate (baseline 12.1%); share of returns with a reason code (target 100%).
- **Confidence and limitations:** High confidence the gap is real. Causes are **hypotheses only** (product descriptions, size expectations, damage in transit, quality); none can be tested here. External context [S1]: NRF estimates 19.3% of online sales are returned, so furniture is below the industry average; the issue is that it is the *worst performer relative to this store's own catalog*. Benchmark definitions differ (NRF: retailer-estimated share of sales, all categories).

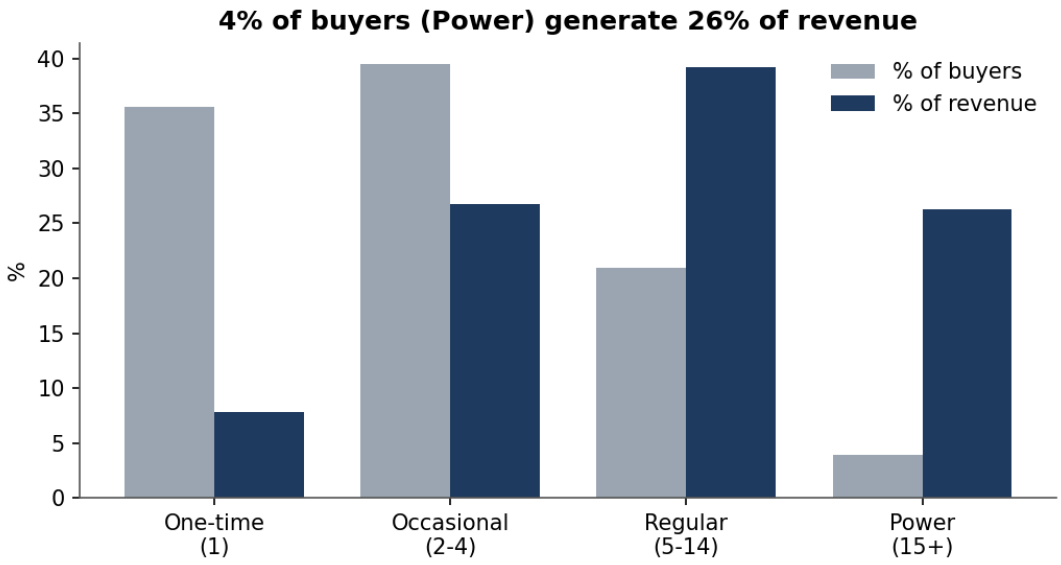
5.3 Paid social acquires customers who rarely buy



Channel	Customers	Bought	Repeat (of buyers)	Activated in 90d	Revenue / customer	Revenue share
referral	367	67.6%	73.4%	56.6%	$747	14.6%
organic_search	1,230	60.2%	66.5%	49.1%	$629	41.1%
email	423	64.1%	66.1%	52.9%	$572	12.9%
paid_search	550	59.5%	62.4%	51.1%	$528	15.4%
affiliate	199	46.7%	64.5%	40.0%	$319	3.4%
paid_social	831	44.2%	54.5%	33.2%	$287	12.7%

- **What the data shows:** Paid social brought 831 customers (23.1%) who generated $287 each versus the $523 company average. Buyer rate across channels differs significantly (chi-square p = 3e-19).
- **Evidence:** Robust to timing: signup-year mix is almost identical across channels, and paid social is lower *within* each cohort (2024 signups: 54.8% bought vs 76.7% elsewhere; 2025 signups: 36.3% vs 50.1%; both p < 0.001). Absolute buyer rates for 2025 signups are understated for every channel because those customers have had little time; only the comparison is reliable. Affiliate looks similarly weak (47% bought) but is small (199 customers).
- **Why it matters:** If paid social costs about the same per signup as other channels, each dollar buys far less revenue. **Spend is not in the dataset, so profitability cannot be concluded.** Referral is the best-quality channel ($747 per customer, 68% bought, 73% repeat) but supplies only 10.2% of customers.
- **Recommended action:** Obtain channel spend and compute CAC and payback; if paid social CAC is not materially lower, shift budget in a controlled test. Trial a referral incentive to scale the best channel.
- **Priority:** High if spend data confirms; otherwise Medium.
- **Measurement:** Buyer rate and 90-day activation by channel; revenue per customer by signup cohort; CAC payback.
- **Confidence:** High on the quality gap; low on the financial conclusion (no cost data).

5.4 A small group of customers carries the business, and many registrants never buy



- **What the data shows:** The top 10% of buyers generate 44.2% of revenue and the top 20% generate 62.0%. 81 Power customers (15+ completed orders; 4.0% of buyers) account for 26.2% of revenue. 729 buyers (35.6%) purchased once and contribute 7.8%.
- **Evidence:** 1,494 of 3,600 registered customers (41.5%) never placed an order; in cohorts with a full 90-day window only 46% order within 90 days of signup. 64% of buyers purchase at least twice.
- **Why it matters:** Revenue depends on retaining a few heavy buyers, while a large pool of registrants is not converting. If the Power group is business or reseller accounts (not identifiable here), their behaviour is closer to B2B and should be managed accordingly.
- **Recommended action:** Quick win: a welcome and activation journey for non-buyers; medium term: early access or service perks for Regular and Power customers; check whether Power accounts are resellers.
- **Priority:** Medium.

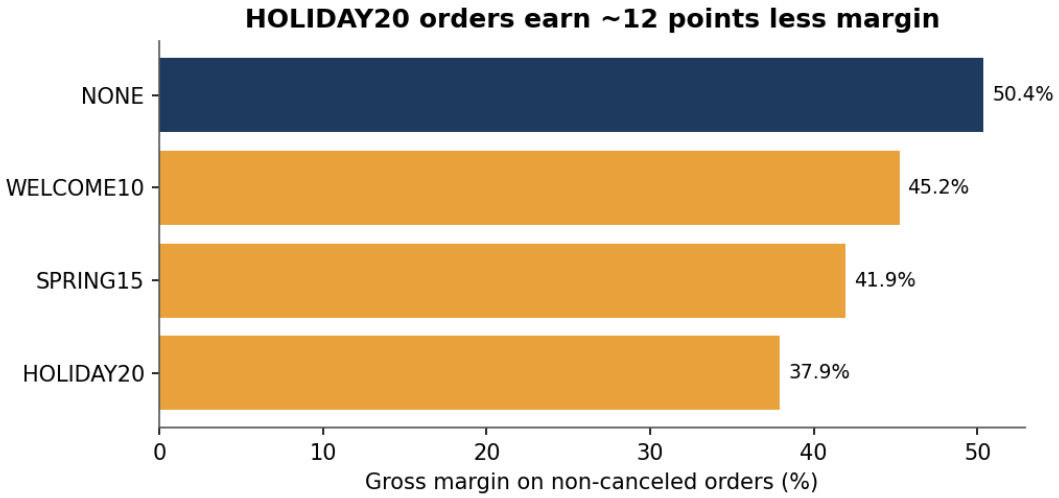
- **Measurement:** 90-day activation rate (baseline 46%); share of revenue from top decile; 12-month retention of Regular and Power segments.
- **Confidence:** High on concentration; the reason for non-purchase is a hypothesis.

5.5 2025 growth rests on adding customers, with weaker retention behind it

	2024	2025
New signups (calendar year)	1,218	2,202
Active buyers (completed order in year)	849	1,626
Orders per active buyer	4.55	2.82
Revenue per active buyer	$1,002	$634

- **What the data shows:** Active buyers grew 92% while revenue per active buyer fell 37%. 50% of 2024 buyers bought again in 2025 and generated 39% of 2025 revenue.
- **Interpretation:** Part of the per-buyer decline is mechanical: 2025 buyers are newer and have had less time to reorder. Even so, growth is clearly tied to continued acquisition. **This is a descriptive finding, not a forecast.**
- **Recommended action:** Track retention by signup cohort alongside revenue growth; treat the 2025 retention rate as a baseline.
- **Priority:** Medium. **Measurement:** 12-month repeat rate by cohort; revenue from returning vs new customers.

5.6 Promotions cost margin; HOLIDAY20 is aimed at existing customers and its incremental effect is untested



Code	Orders	Avg discount	Total discount	Margin	Share on first order
HOLIDAY20	748	$42.78	$32,001	37.9%	22.5%
WELCOME10	373	$22.46	$8,377	45.2%	27.3%
SPRING15	233	$32.25	$7,515	41.9%	21.5%
NONE	7,509	$0.00	$0	50.4%	23.0%

- **What the data shows:** HOLIDAY20 accounts for $32,001 (67%) of all code discounts and is used on 35% of Nov-Dec orders; only 22.5% of its orders are first orders (baseline 23.0%), so it is not an acquisition tool. WELCOME10 should be a new-customer code, yet only 27.3% of its uses are first orders: $6,051 of $8,377 went to repeat orders, and 52 customers used it more than once (up to 7 times).
- **Evidence on seasonality:** Nov-Dec average monthly revenue ran about 41% (2024) and 37% (2025) above other months. Because 65% of Nov-Dec orders used no code, the seasonal lift cannot be attributed to the code.
- **Interpretation:** The lower margin on coded orders is mechanical (the discount itself); the real question is *incrementality*: how many HOLIDAY20 buyers would have purchased anyway. The data cannot answer this; there is no control group.

- **Recommended action:** Run a holdout test for the next holiday campaign (e.g., a random share of eligible customers sees no code or 10%); enforce one use per customer on WELCOME10 (quick win; small dollar value).
- **Priority:** Medium (HOLIDAY20 test); Low (WELCOME10 leakage, $6,051 over two years).
- **Measurement:** Incremental orders and margin per promoted customer vs holdout; discount as % of gross sales (baseline 2.3% on completed orders).
- **Confidence:** High on the descriptive numbers; incrementality unknown. External context [S3]: US online holiday spending reached $257.8B in Nov-Dec 2025 (+6.8% YoY), so some of the seasonal peak reflects market-wide demand.

5.7 Tested and found nothing (reported to prevent wasted effort)

- **Free-shipping threshold:** Orders just below $75 (n = 335) and just above (n = 352) are similar in number (ratio 1.05, p = 0.54); no evidence that the threshold pushes customers to add items.
- **Cancellations by payment method:** Rates range from 3.4% to 4.4% with no significant difference (p = 0.15).
- **Product concentration:** 51 of 90 products produce 80% of revenue and the top 10 only 24%, so there is no single-SKU dependency.
- **Geography:** Top-5 states hold 42% of revenue; per-customer figures in small states are driven by a few heavy buyers and were not interpreted.

6. Business implications

Growth is healthy but its quality needs watching: customer acquisition is doing the heavy lifting, one acquisition channel converts poorly, one category leaks revenue through returns, and promotions are not yet proven to add demand. None of these is a crisis; each has a low-cost diagnostic step.

7. Recommendations and roadmap

Horizon	Action	Evidence	Priority
Quick win (0-30 days)	Capture return reason codes (all categories, furniture first)	5.2	High
Quick win	Pull channel spend; compute CAC and payback for paid social, affiliate, referral	5.3	High
Quick win	Limit WELCOME10 to one use per customer	5.6	Low
Medium (30-90 days)	Audit the five highest-return furniture products	5.2	High
Medium	Activation journey for registrants with no order; test referral incentive	5.3, 5.4	Medium
Medium	Cohort retention dashboard	5.5	Medium
Longer term	HOLIDAY20 holdout test before next Nov	5.6	Medium
Longer term	Maintain cost history so margin trends are interpretable	5.1	Medium

8. KPI monitoring plan

KPI	Baseline (this data)	Source	Review
Furniture line return rate	12.1%	fact_items	Monthly
Return reason code coverage	0% (field absent)	to be captured	Monthly
Paid social buyer rate / revenue per customer	44.2% / $287	customers + orders	Quarterly
90-day activation	46.4%	customers + orders	Monthly
12-month repeat rate of 2024 buyers	50%	orders	Quarterly
Discount % of gross sales	2.3%	order_items	Monthly
Share of revenue from top 10% of buyers	44.2%	orders	Quarterly

9. Limitations and assumptions

- Returned orders recognise zero revenue and their cost is assumed recovered (restocked). If goods are not restocked, losses are understated.
- Return status is per order, so partial returns are invisible.

- Gross profit ignores shipping cost, payment fees, returns handling and marketing.
- `unit_cost` has no history; 2025 margin gains are likely mechanical.
- Heavy customers (15+ orders) could be resellers; results were re-tested without them where relevant.
- Dataset provenance and licence are not documented; no external validation of the data itself is possible.
- All causes given for patterns are hypotheses. Nothing here is a forecast.

10. Sources

External evidence is listed in `reports/sources.md` and labelled [S1]-[S3] above. **[S1]** NRF & Happy Returns, 2025 Retail Returns Landscape (15 Oct 2025). **[S2]** US Census Bureau, Quarterly Retail E-Commerce Sales (10 Mar 2026). **[S3]** Adobe Analytics 2025 holiday results.

Appendix: reproducibility

```
python run_pipeline.py && python -m src.sql_reconcile && python -m pytest && python -m src.build_reports
```